Dataset	Efficientnet-b7 + SAM (optimal)	Efficientnet-b7 + SAM ($\rho = 0.05$)	Efficientnet-b7
FGVC_Aircraft	6.80	7.06	8.15
Flowers	0.63	0.81	1.16
Oxford_IIIT_Pets	3.97	4.15	4.24
Stanford_Cars	5.18	5.57	5.94
CIFAR-10	0.88	0.88	0.95
CIFAR-100	7.44	7.56	7.68
Birdsnap	13.64	13.64	14.30
Food101	7.02	7.06	7.17

Table 10: Results for the finetuning experiments, using $\rho = 0.05$ for all datasets.

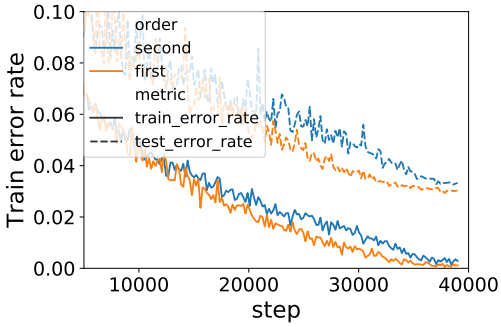

Figure 4: Training and test error for the first and second order version of the algorithm.

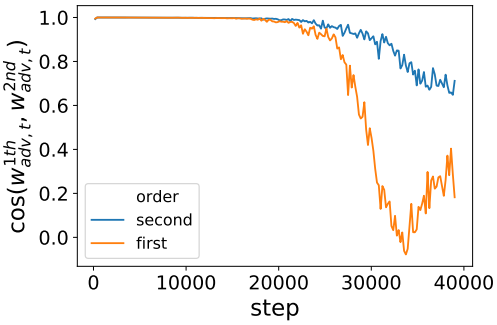

Figure 5: Cosine similarity between the first and second order updates.

with the test accuracy and the estimated sharpness ($\max_{\boldsymbol{\epsilon}} L(\boldsymbol{w} + \boldsymbol{\epsilon}) - L(\boldsymbol{w})$) at the end of training in Table 11; we report means and standard deviations across 20 runs.

For most of the training, one projected gradient step (as used in standard SAM) is sufficient to obtain a good approximation of the $\boldsymbol{\epsilon}$ found with multiple inner maximization steps. We however observe that this approximation becomes weaker near convergence, where doing several iterations of projected gradient ascent yields a better $\boldsymbol{\epsilon}$ (for example, on CIFAR-10, the maximum loss found on each batch is about 3% more when doing 5 steps of inner maximization, compared to when doing a single step). That said, as seen in Table 11, the test accuracy is not strongly affected by the number of inner maximization iterations, though on CIFAR-100 it does seem that several steps outperform a single step in a statistically significant way.

Number of projected gradient steps	CIFAR-10		CIFAR-100	
	Test error	Estimated sharpness	Test error	Estimated sharpness
1	$2.77_{\pm 0.03}$	$0.17_{\pm 0.03}$	$16.72_{\pm 0.08}$	$0.82_{\pm 0.05}$
2	$2.76_{\pm 0.03}$	$0.82_{\pm 0.03}$	$16.59_{\pm 0.08}$	$1.83_{\pm 0.05}$
3	$2.73_{\pm 0.04}$	$1.49_{\pm 0.05}$	$16.62_{\pm 0.09}$	$2.36_{\pm 0.03}$
5	$2.77_{\pm 0.03}$	$2.26_{\pm 0.05}$	$16.60_{\pm 0.06}$	$2.82_{\pm 0.04}$

Table 11: Test error rate and estimated sharpness ($\max_{\epsilon} L(w+\epsilon) - L(w)$) at the end of the training.

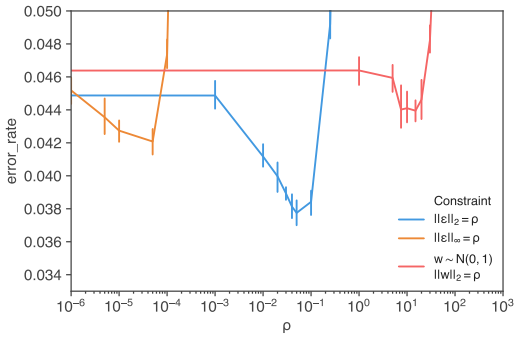

Figure 6: Test accuracy for a WideResNet trained on CIFAR-10 with SAM, for different perturbation norms.

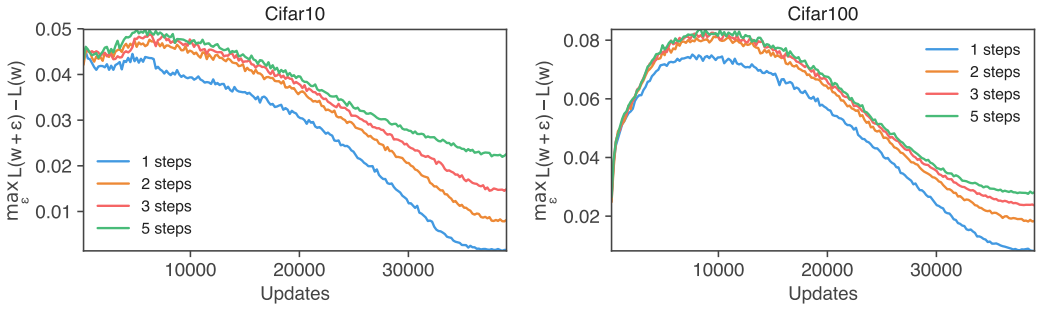

Figure 7: Evolution of $\max_{\boldsymbol{\epsilon}} L(\boldsymbol{w} + \boldsymbol{\epsilon}) - L(\boldsymbol{w})$ vs. training step, for different numbers of inner projected gradient steps.